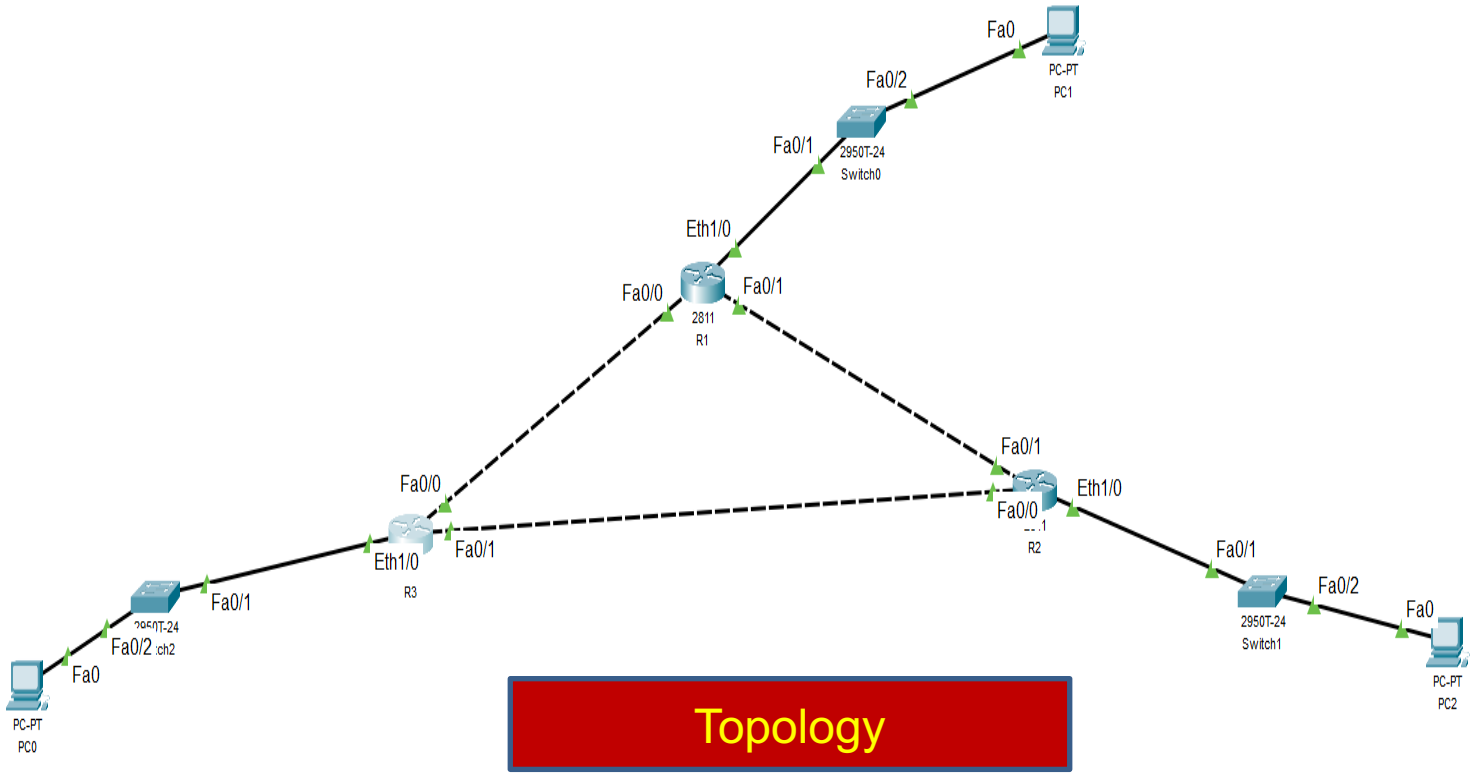
Cisco Packet Tracer - C:\Users\Gautham\Cisco Packet Tracer 8.2.2\saves\pkt2.pkt

File Edit Options View Tools Extensions Window Help

Logical Physical x: 850, y: 387 Root 09:41:30




Time: 00:19:01 Realtime Simulation

4331 4321 1941 2901 2911 819IOX 819HGW 829 1240 PT-Router PT-Empty 1841

829

Scenario 0

New Delete

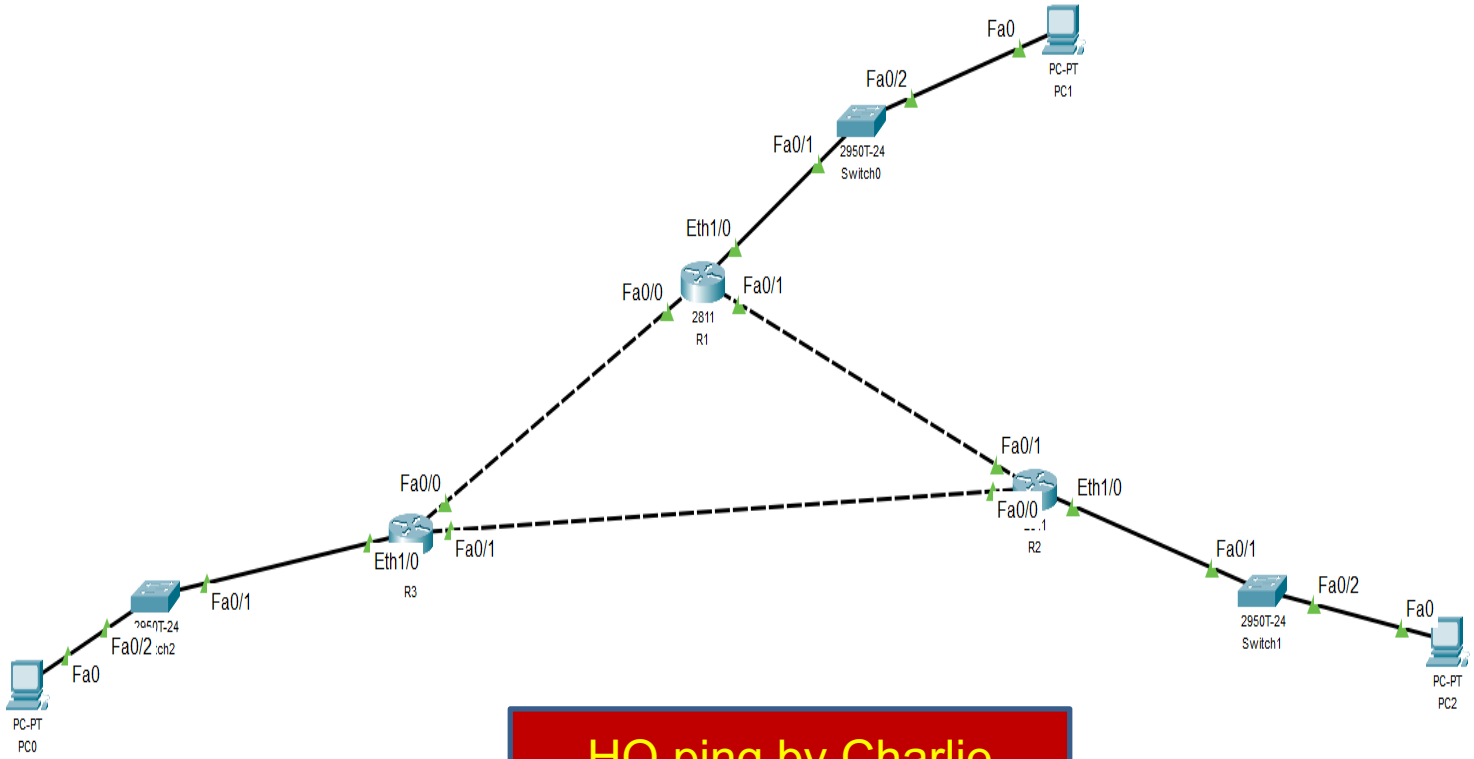
Toggle PDU List Window

| Fire | Last Status | Source | Destination | Type | Color | Time(sec) | Periodic | Num | Edit | Delete |
|---|---|---|---|---|---|---|---|---|---|---|
| | Failed | PC1 | PC2 | ICMP | | 0.000 | N | 0 | (edit) | (delete) |

Cisco Packet Tracer - C:\Users\Gautham\Cisco Packet Tracer 8.2.2\saves\pkt2.pkt

File Edit Options View Tools Extensions Window Help

Logical | Physical x: 1033, y: 338 | Root | 10:14:00




HQ ping by Charlie

Time: 00:20:05 | Realtime | Simulation

| Fire | Last Status | Source | Destination | Type | Color | Time(sec) | Periodic | Num | Edit | Delete |
|---|---|---|---|---|---|---|---|---|---|---|
| | Successful | PC0 | PC1 | ICMP | | 0.000 | N | 0 | (edit) | (delete) |

Scenario 0 | New | Delete | Toggle PDU List Window

4331 4321 1941 2901 2911 819IOX 819HGW 829 1240 PT-Router PT-Empty 1841 2620

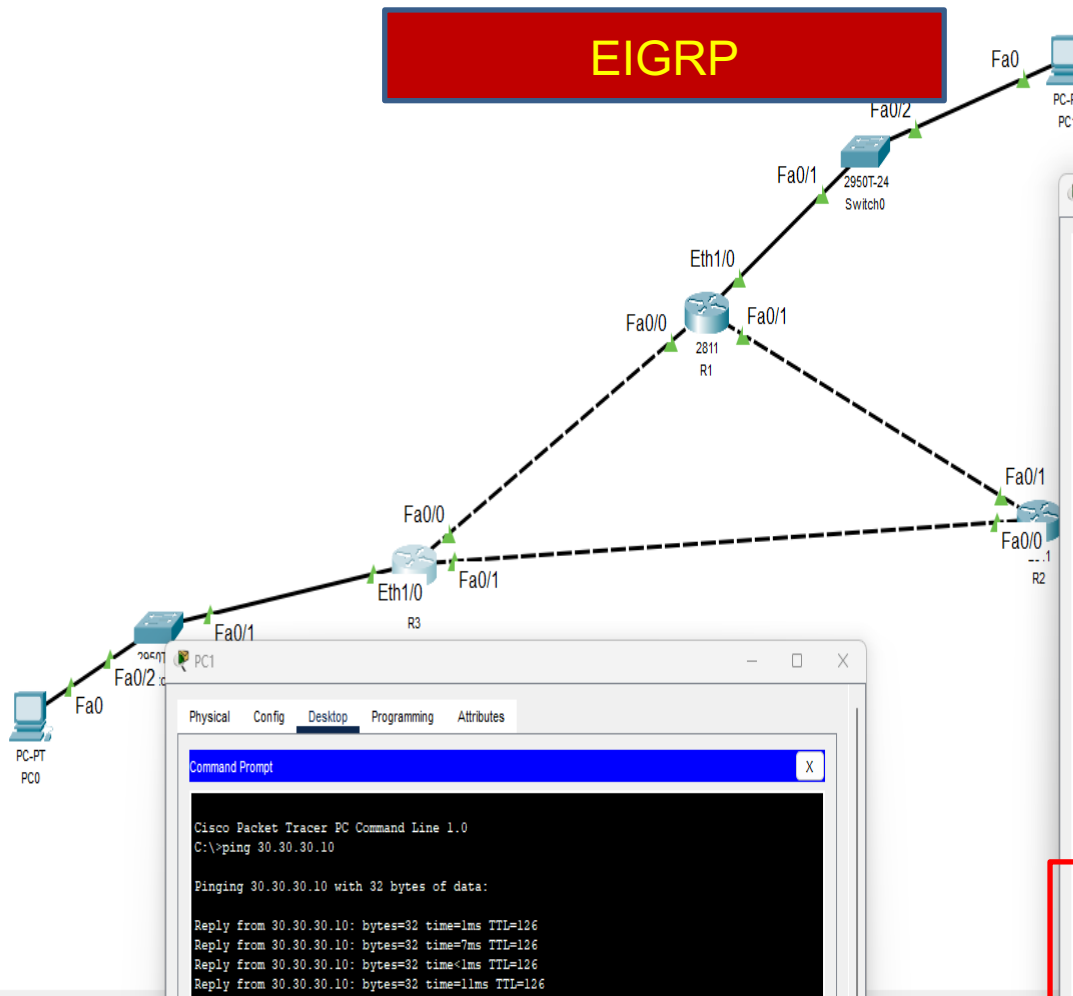
829

Cisco Packet Tracer - C:\Users\Gautham\Cisco Packet Tracer 8.2.2\saves\pkt2.pkt

File Edit Options View Tools Extensions Window Help

Logical | Physical x: 841, y: 465 | Root | 09:23:00

# EIGRP




PC1

Physical | Config | Desktop | Programming | Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 30.30.30.10

Pinging 30.30.30.10 with 32 bytes of data:

Reply from 30.30.30.10: bytes=32 time=1ms TTL=126
Reply from 30.30.30.10: bytes=32 time=7ms TTL=126
Reply from 30.30.30.10: bytes=32 time<1ms TTL=126
Reply from 30.30.30.10: bytes=32 time=11ms TTL=126

Ping statistics for 30.30.30.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 11ms, Average = 4ms

C:\>
```

R3

Physical | Config | CLI | Attributes

GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
FastEthernet0/0
FastEthernet0/1
Ethernet1/0
Ethernet1/1
Ethernet1/2
Ethernet1/3

Global Settings

Display Name: R3
Hostname: Router
NVRAM: Erase | Save
Startup Config: Load... | Export...
Running Config: Export... | Merge...

Equivalent IOS Commands

```
Ethernet1/1            unassigned      YES unset  administratively down down
Ethernet1/2            unassigned      YES unset  administratively down down
Ethernet1/3            unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 192.168.13.2 255.255.255.0
Router(config-if)#
%DUAL-5-NBRCHANGE: IP-EIGRP 100: Neighbor 192.168.13.1 (FastEthernet0/0) is up: new
adjacency
```

Time: 00:18:25

Destination | Type | Color | Time
PC2 | ICMP

R3

Physical Config CLI Attributes

IOS Command Line Interface

```
2 FastEthernet interface(s)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!


%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/0, changed state to up

%DUAL-5-NBRCHANGE: IP-EIGRP 100: Neighbor 192.168.13.1 (FastEthernet0/0) is up: new adjacency


Router>enable
Router#ip route
          ^
% Invalid input detected at '^' marker.

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

D EX 10.0.0.0/8 [170/284160] via 192.168.13.1, 00:03:06, FastEthernet0/0
D EX 20.0.0.0/8 [170/284160] via 192.168.13.1, 00:02:17, FastEthernet0/0
     30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       30.0.0.0/8 is directly connected, Ethernet1/0
L       30.30.30.1/32 is directly connected, Ethernet1/0
D EX 192.168.12.0/24 [170/284160] via 192.168.13.1, 00:03:07, FastEthernet0/0
     192.168.13.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.13.0/24 is directly connected, FastEthernet0/0
L       192.168.13.2/32 is directly connected, FastEthernet0/0
     192.168.23.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.23.0/24 is directly connected, FastEthernet0/1
L       192.168.23.2/32 is directly connected, FastEthernet0/1
 --More--
```

IP route r3

Copy Paste

R1

Physical Config CLI Attributes

IOS Command Line Interface

```
R1>enab
00:00:45: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.23.1 on FastEthernet0/1 from LOADING to FULL,
Loading Done
le
R1#ip route
      ^
% Invalid input detected at '^' marker.

R1#config t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ip route
% Incomplete command.
R1(config)#enable
% Incomplete command.
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console
enable
R1#ip route
      ^
% Invalid input detected at '^

R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/8 is directly connected, Ethernet1/0
L       10.10.10.1/32 is directly connected, Ethernet1/0
O    20.0.0.0/8 [110/11] via 192.168.12.2, 00:03:26, FastEthernet0/1
D    30.0.0.0/8 [90/284160] via 192.168.13.2, 00:04:15, FastEthernet0/0
     192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.12.0/24 is directly connected, FastEthernet0/1
L       192.168.12.1/32 is directly connected, FastEthernet0/1
     192.168.13.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.13.0/24 is directly connected, FastEthernet0/0
L       192.168.13.1/32 is directly connected, FastEthernet0/0
D    192.168.23.0/24 [90/30720] via 192.168.13.2, 00:04:15, FastEthernet0/0
 --More--
```

IP route r1

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Top

R2

Physical Config CLI Attributes

IOS Command Line Interface

```
cisco 2811 (MPC860) processor (revision 0x200) with 60416K/5120K bytes of memory
Processor board ID JAD05190MTZ (4292891495)
4 Ethernet interface(s)
2 FastEthernet interface(s)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!


%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/0, changed state to up

00:00:45: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.13.1 on FastEthernet0/1 from LOADING to FULL,
Loading Done


R2>enable
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

O    10.0.0.0/8 [110/11] via 192.168.12.1, 00:04:03, FastEthernet0/1
     20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.0.0.0/8 is directly connected, Ethernet1/0
L       20.20.20.1/32 is directly connected, Ethernet1/0
O E2 30.0.0.0/8 [110/20] via 192.168.12.1, 00:04:03, FastEthernet0/1
     192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.12.0/24 is directly connected, FastEthernet0/1
L       192.168.12.2/32 is directly connected, FastEthernet0/1
O    192.168.13.0/24 [110/2] via 192.168.12.1, 00:04:03, FastEthernet0/1
     192.168.23.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.23.0/24 is directly connected, FastEthernet0/0
L       192.168.23.1/32 is directly connected, FastEthernet0/0
 --More--
```

IP route r2

Copy Paste

Top

R1

Physical | Config | CLI | Attributes

IOS Command Line Interface

```
%DUAL-5-NBRCHANGE: IP-EIGRP 100: Neighbor 192.168.13.2 (FastEthernet0/0) is down:
Interface Goodbye received

%DUAL-5-NBRCHANGE: IP-EIGRP 100: Neighbor 192.168.13.2 (FastEthernet0/0) is up: new
adjacency

R1#
%SYS-5-CONFIG_I: Configured from console by console
show ip ospf neighbor

R1#show ip ospf neighbor

R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA
       E1 - OSPF external type 1, E2 - OSPF external
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2
       * - candidate default, U - per-user static route
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/8 is directly connected, Ethernet
L       10.10.10.1/32 is directly connected, Ether
D    30.0.0.0/8 [90/284160] via 192.168.13.2, 00:0
     192.168.12.0/24 is variably subnetted, 2 subn
C       192.168.12.0/24 is directly connected, Fas
L       192.168.12.1/32 is directly connected, Fas
     192.168.13.0/24 is variably subnetted, 2 subn
C       192.168.13.0/24 is directly connected, Fas
L       192.168.13.1/32 is directly connected, FastEthernet0/0
D    192.168.23.0/24 [90/30720] via 192.168.13.2, 00:00:28, FastEthernet0/0

R1#
```

**Missing Network statement on r1**

- Configure terminal router OSPF 1 no network 192.168.12.0 0.0.0.3 area 0 end
- Stops linking to r2 because OSPF stops sending hellos
- Can be restored by configure terminal router OSPF 1 network 192.168.12.0 0.0.0.3 area 0 end

Copy | Paste

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